



ICARUS

Team M2

Table of Contents

- Client / Team
- Project assignment
- Application & AI Design
- Demo
- DevSecOps

Client

Their mission:

- Reduce AI hallucinations
- Deliver reliable AI decisions
- Enable business-ready integrations
- Policy compliant decision making

Team

- Multidisciplinary team
- Roles across APP / AI / CCS

APP: Sybren Backx

AI: Ravin Shalmashi, Paschal Chukwubuikem Ifewulu, Thomas Van Sande , Youssef Nait Salah

CCS:Evert Schol , Maximus Mukiza

Project Assignment

- Extend Cloudway's AI + logic solution
- Enable system-to-system communication
- Build a management platform for integrations
- Easy to Configure integrations
- Logging and audits
- Deployable and scalable cloud solution



Application design

- React typescript + node.js
- PostgreSQL
- Unit Testing
- Code quality SonarQube



Emitted Events

- Persistent Job Queue
- RequestStatus
 - *Pending*
 - *Processing*
 - *Completed*
 - *Failed*
- Automatic Default Retries

EVENTS

MessageReceived

FetchFacts

CallFirstModel

CallLogicUnit

ReadabilityPostProcess

FinalizeRequest

Webhooks

- Configurable endpoints
- Multiple endpoints
- Trigger events

Webhooks

Webhook URL

`https://ddmxqat4n21fj.cloudfront.net/api/webhook-test`

Trigger events

☒ request.completed

☒ request.failed

Remove webhook

AI MODELS

AI Pre-Processing (Input Understanding Layer)

- Convert raw user input into structured JSON
- Extracts relevant facts
- Filters out irrelevant information
- Keeps context across multiple messages in the same session
- Text + images + documents
- Configurable system prompt

Model Selection & Multi-Modal Processing

Each integration defines the pre-processing model:

- Text-only model
- Multi-modal model (text + images + documents)
- Amazon Nova Lite

Output

- A clean JSON structure containing extracted facts
- Used as the single source of truth for the Logic Unit

AI Post-Processing (Readability Layer)

- Transform raw Logic Unit output into a clear, user-friendly result
- Ensure consistency, correctness, and professional formatting
- Preserves the original decision (approved / denied / pending)
- Converts technical output into readable explanations
- Explicitly highlights missing information and next steps
- Never invents facts or alters Logic Unit conclusions

AI Post-Processing

Output formats :

Output Format

Text

Text

JSON

PDF

DOCX

XLSX

JSON Schema:

Output JSON Schema

```
"type": "object",
"additionalProperties": false,
"required": ["decision", "missing_information",
"next_steps"],
"properties": {
  "decision": { "type": "string", "description":
"Final decision or current claim status" },
  "missing_information": {
    "type": "array",
```

Validate schema

Reset to default

✓ Schema is valid

```
?`  
You are a readability layer. Convert the Logic Unit c
```

Rules:

- DO NOT change the decision.
- Do not invent facts.
- Explicitly state missing information.

Output:

- Plain text only.

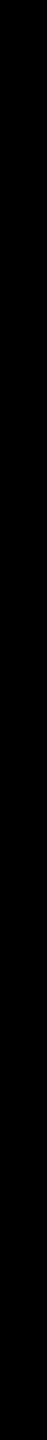
```
`).trim()  
:`
```

```
You are a readability layer. Produce valid JSON only.
```

Schema:

```
{  
  "summary": string,  
  "decision": { "status": "approved" | "denied" | "ne  
  "reasons": string[],  
  "required_actions": string[],  
  "required_documents": string[],  
  "missing_information": string[],  
  "references": { "atoms_used": string[] }  
}  
`).trim();
```

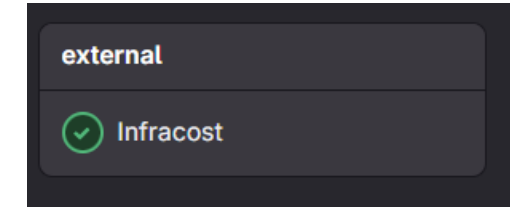
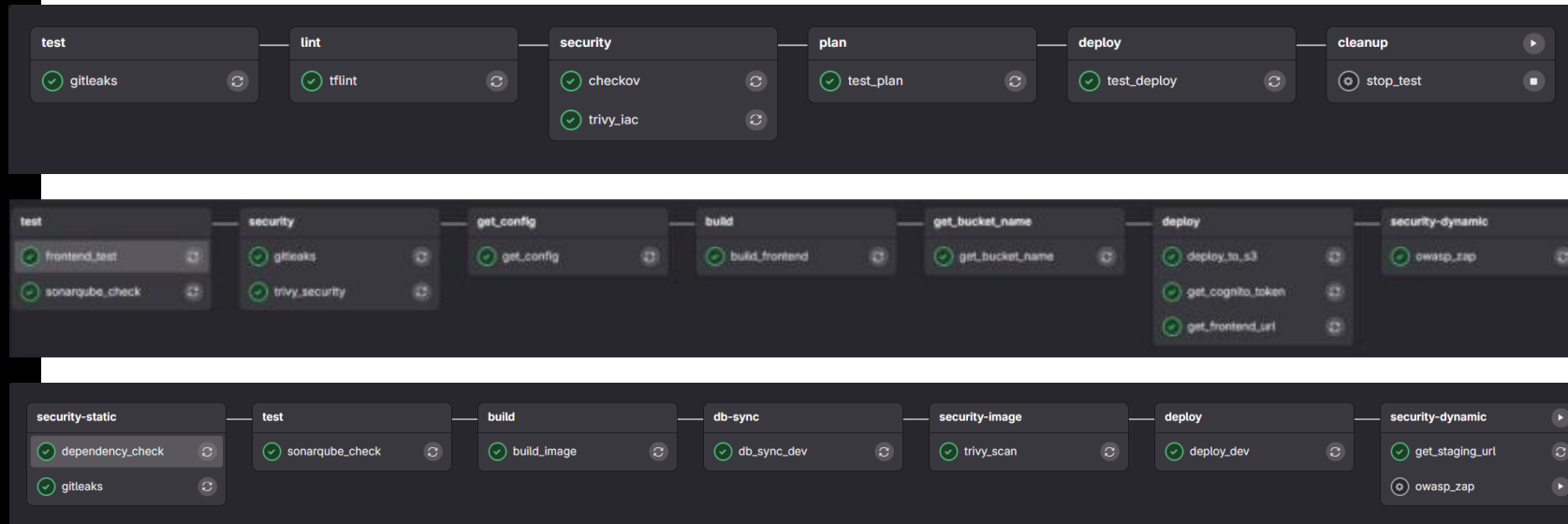
AI POST- PROCESSING – SYSTEM PROMPT



DEMO

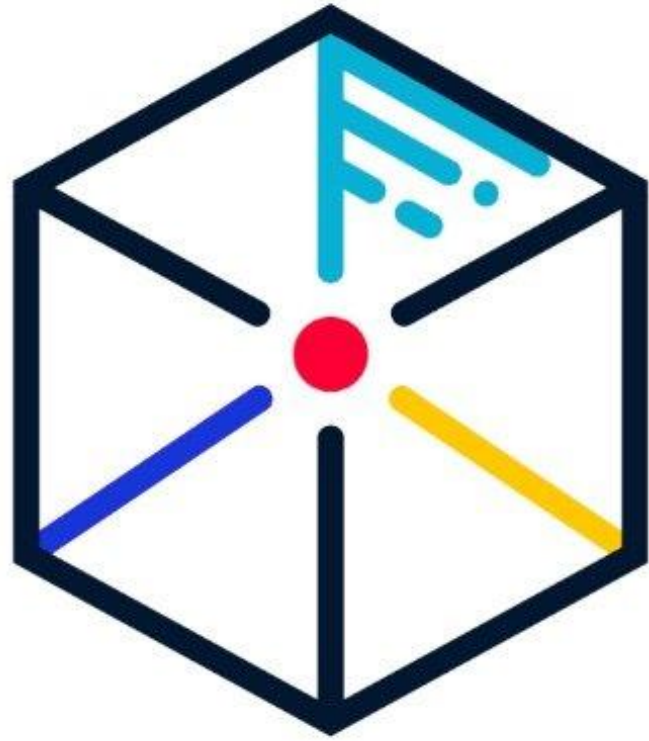
DEVSECOPS





CICD & Gitlab

- Separate pipeline Backend, Frontend, Infrastructure
- Triggers on merge to Test and Main branch
- Zero-downtime deployments to AWS



Security tools

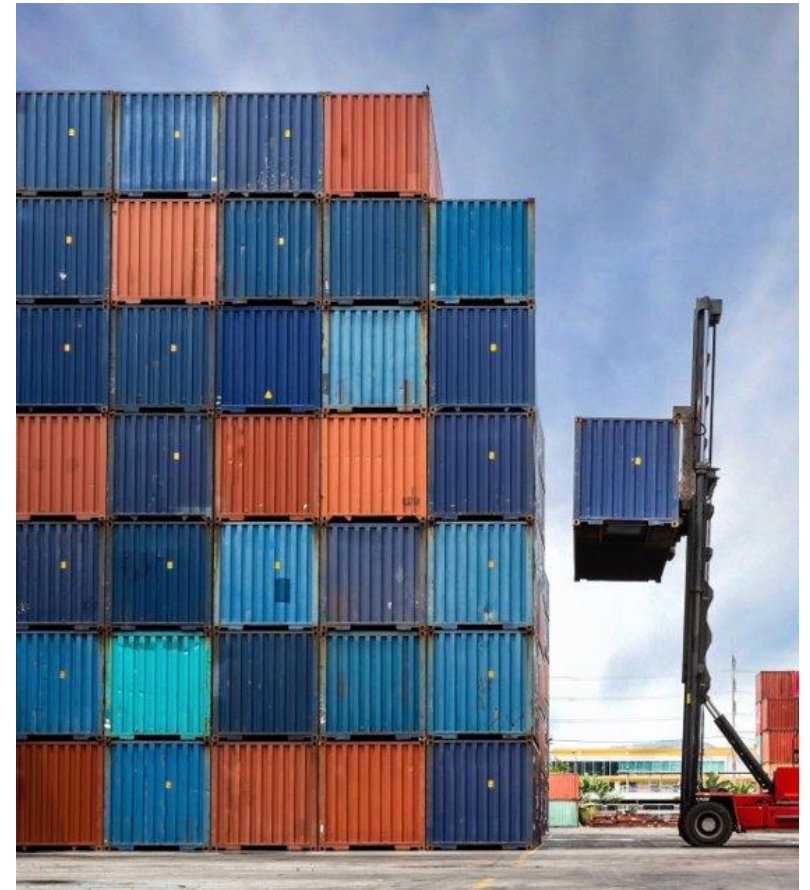
- Gitleaks
- Dependency-Check
- SonarQube
- Trivy
- Checkov & TFLint
- Owasp ZAP



checkov
BY PRISMA® CLOUD

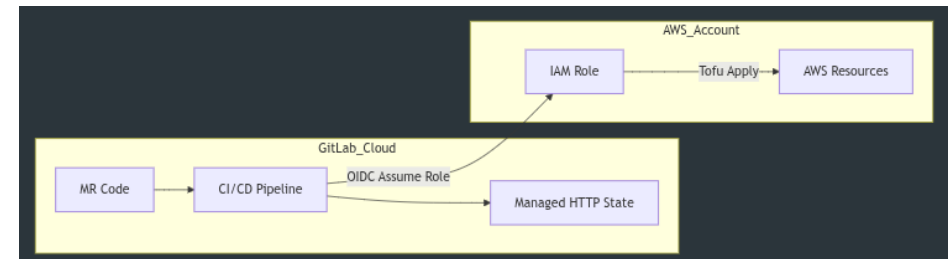
Container

- Multistage build
- Hardened image (chainguard)
- ARM64 for cost reduction



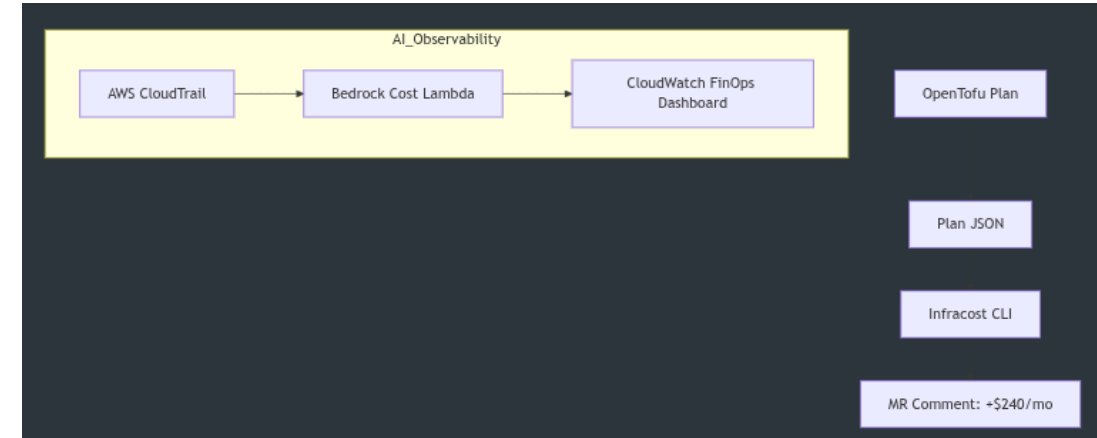
Infrastructure Lifecycle

- Engine: OpenTofu
- State Management: GitLab (HTTP backend)
- Authentication: OpenID Connect (OIDC)
- State locking prevents Ghost Resources.
- Infrastructure lifecycle is decoupled from the local machine configurations.

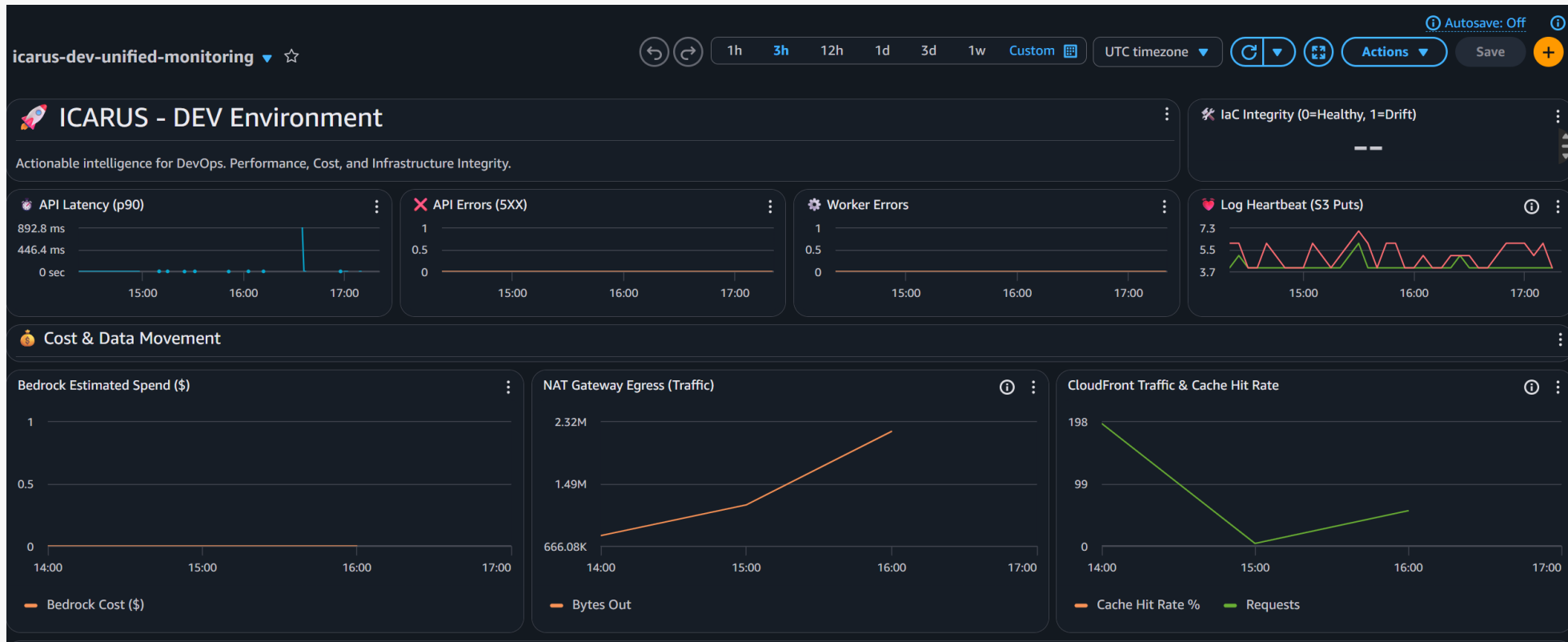


FinOps

- Infracost Integration automatically identifies resource change and monthly cloud spend
- AI token monitoring: custom lambda script for cost per token
- Resource Labeling: enforcement of Cost Center, Services and environment tags.



Dashboard



REFLECTION



Strong points / wow-effect

- Reduced AI hallucinations
- Logic controlled outputs
- Non-technical configuration

Points for improvement

- Logging screen and table
- Audits/human flagging system
- Better UI for our webhook output

Retrospective Teamwork

Technical skills evolution:

- Working in TypeScript
- Combining APP/AI/CCS
- Webhooks
- Asynchronous event-based worker flow
- Being mindful of security best practices
- Multi-modal processing

Retrospective Teamwork

Soft skills evolution:

- Priority on communication
- Stopped working in silos and helped each other out
- Emphasis on early planning and outlining
- Learning what to prioritize
- Learning how to present and prepare for demo meetings

CONCLUSION

